

Predetermined-to-Anytime Transfer for Gradient Descent

A Suffix-Local Addendum

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Abstract

We record the precise transfer property connecting two lower-bound regimes for last-iterate gradient descent on smooth convex functions. By class inclusion alone, a generic finite-horizon minimax ceiling transfers to an anytime ceiling at the same exponent. The stronger map

$$\mathcal{T}(p) = \frac{2p}{1+p}$$

requires more information: the same finite-horizon hard-instance certificate must localize to every relevant ranked suffix at record prefixes. We give a self-contained suffix-local compiler theorem and its near/far proof. The currently audited joint-router certificate satisfies the compiler interfaces, so one computer-assisted package proves simultaneously

$$p_{\text{minimax}} \leq \frac{165737}{100000} = 1.65737, \quad q_{\text{any}} \leq \frac{331474}{265737} = 1.2473761651557743 \dots$$

These are supremal-exponent ceilings. We do not claim endpoint attainment, a positive uniform constant at either endpoint, or an endpoint not- o statement.

1 Two schedule models

Let $\mathcal{R}_n(h_0, \dots, h_{n-1})$ be the normalized worst-case last-iterate objective gap of gradient descent over convex functions with one-Lipschitz gradient. All stepsizes below are nonnegative and fixed before oracle information is observed.

In the predetermined-horizon model, the horizon is announced first and a different finite schedule may be chosen for each horizon. Define

$$p_{\text{minimax}} := \sup \left\{ p \geq 0 : \begin{array}{l} \text{there are } C < \infty \text{ and schedules } h^{(T)} \text{ such that} \\ \mathcal{R}_T(h^{(T)}) \leq C(T+1)^{-p} \text{ for every } T \geq 1 \end{array} \right\}.$$

In the anytime model, one infinite schedule must work at every sufficiently late deterministic prefix:

$$q_{\text{any}} := \sup \left\{ q \geq 0 : \begin{array}{l} \text{there are one infinite schedule } h, C < \infty, \text{ and } n_0 \\ \text{such that } \mathcal{R}_n(h_0, \dots, h_{n-1}) \leq C(n+1)^{-q} \\ \text{for every } n \geq n_0 \end{array} \right\}.$$

The anytime class is more constrained because its finite prefixes must be compatible with one another.

Proposition 1.1 (Black-box prefix transfer). *If a universal finite-horizon theorem proves $\mathcal{R}_T(g) \geq c(T+1)^{-p}$ for every length- T schedule g , then every infinite schedule obeys the same lower bound at every prefix. Hence $q_{\text{any}} \leq p$.*

Proof. Apply the finite-horizon theorem to the first T steps of the infinite schedule. No compatibility beyond taking a prefix is needed. $\square$

Proposition 1.1 does *not* yield the smaller exponent $2p/(1+p)$. That stronger conclusion uses internal, suffix-local data from the hard-instance proof.

2 The suffix-local transfer compiler

Fix $0 < a < 1$ and put

$$p := 1 + a, \quad b := \frac{2}{a}, \quad \Theta_* := \frac{1}{a(a+2)} = \frac{1}{p^2 - 1}, \quad \mathcal{T}(p) := \frac{2p}{1+p} = \frac{2(1+a)}{2+a}. \quad (1)$$

For a finite schedule, write its positive excesses in decreasing order as $c_1 \geq \dots \geq c_r > 0$, and define

$$B := 1 + \sum_{t < n} \min\{h_t, 1\}, \quad D_j := B + \sum_{i=j+1}^r c_i. \quad (2)$$

Thus $D_0 = 1 + \sum_{t < n} h_t$, $D_r = B$, $B \leq n + 1$, and $r \leq n$.

Theorem 2.1 (Suffix-local transfer). *Suppose one fixed chronological ranked-path certificate has the following properties uniformly at every deterministic prefix.*

1. *The empty-chain and strict-record hard instances give*

$$\mathcal{R}_n \geq \frac{1}{2D_0 - 1}, \quad c_1 \leq D_0 \sqrt{\mathcal{R}_n} \quad (3)$$

at every prefix whose last step is a strict record exceeding one.

2. *For every $2 \leq q \leq r$, all top- q excesses and all $q - 1$ chronological edges occur in a hard-instance bound*

$$\mathcal{C}_n(h) \geq \frac{c_J}{D_q} \exp \left\{ \mu(z + q - S_v) + \sum_{i=1}^{q-1} \Delta(v_i, v_{i+1}) \right\}, \quad (4)$$

where $c_J > 0$, $0 < \mu \leq 1$, $z \geq 0$, $S_v = \sum_i v_i$, and (v_i) is a chronological permutation of

$$\hat{v}_i = \frac{2D_q}{qc_i}, \quad 1 \leq i \leq q.$$

The normalized worst-case risk satisfies $\mathcal{R}_n(h) \geq \mathcal{C}_n(h)$.

3. *Fix finite thresholds $0 < t_0 < \dots < t_{M-1} < \infty$, and set*

$$\alpha_\ell := \frac{1}{q} \#\{i : v_i \leq t_\ell\}, \quad k_\ell := q\alpha_\ell,$$

$$\mathcal{R}_\ell(\alpha) := 1 + 2 \sum_{u=\ell}^{M-2} \frac{\alpha_{u+1} - \alpha_u}{t_u} + \frac{2(1 - \alpha_{M-1})}{t_{M-1}}.$$

The exact denominator router, including zero occupancy, satisfies

$$\frac{D_{k_\ell}}{D_q} \leq \mathcal{R}_\ell(\alpha).$$

Every state row with positive dual weight is either a cumulative occupancy monotonicity row or a globally valid affine tangent row

$$\mathcal{R}_{\ell(m)}(\alpha) \geq T_m(\alpha_{\ell(m)}), \quad T_m(x) \leq x^{-a} \quad (0 < x \leq 1).$$

There is no positive-weight mean row or other state row, and

$$H := \max_m \sup_{0 \leq x \leq 1} T_m(x) < \infty.$$

4. There are finite Q and B_0 such that every supported state with $q \geq Q$ satisfies the signed finite-rank estimate

$$\frac{1}{q} \sum_{i=1}^{q-1} \Delta(v_i, v_{i+1}) \geq B_0.$$

This estimate includes all physical edges, endpoint corrections, and the unbounded state group. Its endpoint reserve

$$\mathfrak{r} := B_0 - \mu(2\Theta_* - 1)$$

is positive. There is a finite localization factor $\Lambda > 1$ with

$$\Lambda^a \geq H, \quad \delta := \mathfrak{r} - \frac{\mu b}{\Lambda} > 0. \quad (5)$$

Then no infinite predetermined schedule satisfies $\mathcal{R}_n = O(n^{-s})$ for any $s > \mathcal{T}(p)$. Consequently,

$$\boxed{q_{\text{any}} \leq \mathcal{T}(p) = \frac{2p}{1+p}.} \quad (6)$$

Proof. Assume that one infinite schedule satisfies $\mathcal{R}_n \leq An^{-s}$ for some $s > \mathcal{T}(p) > 1$. The empty-chain arm rules out bounded steps, so there are arbitrarily late strict-record prefixes. At such a prefix write $\varepsilon = \mathcal{R}_n$, choose

$$\eta = \frac{1}{4\Lambda}, \quad k = \left\lfloor \frac{\eta}{\sqrt{\varepsilon}} \right\rfloor,$$

and let q minimize $P_j := D_j j^a$ over $k \leq j \leq r$. For late records, $k \geq \eta/(2\sqrt{\varepsilon})$. The record arms imply

$$D_m \geq (1 - \eta)D_0 \geq \frac{1 - \eta}{2\varepsilon} \quad (0 \leq m \leq \min\{k, r\}). \quad (7)$$

In particular, eventually $r \geq k$. Indeed, if $r < k$, then (7) with $m = r$ gives $B = D_r \geq c/\varepsilon$, contradicting $B \leq n + 1$, $\varepsilon \leq An^{-s}$, and $s > 1$.

If $q < \Lambda k$, then $D_q \geq 3D_0/4$, while suffix minimality and $B \leq n + 1$, $r \leq n$ give

$$c\varepsilon^{-(p+1)/2} \leq D_k k^a \leq \frac{4}{3} D_q q^a \leq \frac{4}{3} B r^a \leq \frac{4}{3} (n + 1)^p.$$

Therefore

$$\varepsilon \geq c'(n + 1)^{-2p/(1+p)},$$

contradicting $s > \mathcal{T}(p)$.

Suppose instead that $q \geq \Lambda k$, and put $\tau = k/q \leq 1/\Lambda$. Suffix minimality first gives the boundary floor $0 < \widehat{v}_i \leq b$. It also validates every supported tangent row. Indeed, if $\alpha_\ell \geq \tau$, then $k_\ell \geq k$ and

$$\mathcal{R}_\ell(\alpha) \geq \frac{D_{k_\ell}}{D_q} \geq \left(\frac{q}{k_\ell} \right)^a = \alpha_\ell^{-a} \geq T_m(\alpha_\ell).$$

If $\alpha_\ell < \tau$, then $k_\ell < k$ and

$$\mathcal{R}_\ell(\alpha) \geq \frac{D_{k_\ell}}{D_q} \geq \frac{D_k}{D_q} \geq \left(\frac{q}{k} \right)^a \geq \Lambda^a \geq H \geq T_m(\alpha_\ell).$$

The latter argument includes $\alpha_\ell = 0$, so no 0^{-a} is evaluated. Suffix minimality also yields weak reciprocal supermajorization against $b((i - 1/2)/q)^p$. Convex midpoint quadrature then controls the actual full-path mean without deleting a vertex or edge:

$$S_v \leq q \left(2\Theta_* + \frac{b}{\Lambda} \right). \quad (8)$$

Since $q \geq \Lambda k \geq c\epsilon^{-1/2} \geq c'n^{s/2}$, eventually $q \geq Q$. Item 4 and equations (4), (5), and (8) imply

$$\mathcal{C}_n(h) \geq c_J \frac{e^{\delta q}}{D_q}. \quad (9)$$

Since $D_q q^a \leq Br^a \leq (n+1)^p$, the assumed risk upper bound and (9) force $q = O(\log n)$. On the other hand, $q \geq \Lambda k \geq c\epsilon^{-1/2} \geq c'n^{s/2}$, a contradiction. The near and far branches prove (6). $\square$

3 Current exact instantiation

Corollary 3.1 (Audited paired ceilings). *The independently audited joint-router certificate at*

$$p = \frac{165737}{100000}, \quad a = \frac{65737}{100000}, \quad \mu = \frac{8161}{10000}$$

satisfies every hypothesis of Theorem 2.1. It has

$$Q = 10^9, \quad B_0 = \frac{118264177243}{10^{12}}, \quad H = \frac{63159358713716857163}{1250000000000000000} < 60.$$

Taking $\Lambda = 10^6$, the remaining exact suffix reserve is

$$\delta = \frac{133808620593933067}{17468753169000000000000} > 0.$$

$$p_{\minimax} \leq \frac{165737}{100000} = 1.65737, \quad q_{\text{any}} \leq \frac{331474}{265737} = 1.2473761651557743 \dots \quad (10)$$

The known silver constructions give the complementary achievable exponents. Writing $s_{\text{sil}} = \log_2(1 + \sqrt{2})$, the current audited research intervals are therefore

$$s_{\text{sil}} \leq p_{\minimax} \leq \frac{165737}{100000}, \quad \frac{2s_{\text{sil}}}{1 + s_{\text{sil}}} \leq q_{\text{any}} \leq \frac{331474}{265737}. \quad (11)$$

Numerically, the constructive anytime endpoint is $2s_{\text{sil}}/(1 + s_{\text{sil}}) = 1.119545204061651 \dots$

Regime	achievable exponent	universal ceiling
Predetermined horizon	$\log_2(1 + \sqrt{2}) = 1.2715533 \dots$	$165737/100000 = 1.65737$
Anytime	$1.119545204061651 \dots$	$331474/265737 = 1.2473761651 \dots$

Table 1: Current independently audited research intervals. Smaller universal ceilings are stronger.

4 Scope and evidence

Theorem 2.1 transfers a *certificate*, not merely a number. In particular, the implication

$$p_{\text{minimax}} \leq p \implies q_{\text{any}} \leq \frac{2p}{1+p}$$

is not valid as a black-box theorem. A finite-horizon induction may compress past the suffix selected at a stopping-time record and therefore fail to provide the local inequality used above. Within the suffix-local certificate class the map is monotone,

$$\mathcal{T}'(p) = \frac{2}{(1+p)^2} > 0,$$

so every certified improvement in p automatically improves the anytime ceiling as well.

The exact $p=1.65737$ verifier, certificate, controlled-repair replay, generic suffix-local compiler, and independent audits all pass. The principal independent audit is

`audit_joint_tilted_common_multiplier_p165737_independent_20260820_v1.md`.

Its clean isolated replay and all transitive manifests pass. The audit independently checks the 199 positive dual rows, all 162,735 pair costs, exact stationarity, the signed potential, finite- Q corrections, the global tangent caps, and the positive suffix reserve displayed above.

This addendum is a research promotion of the paired ceilings, not a rewrite of the hash-pinned $\sqrt{3}$ or 607/366 release reports. It makes no claim that either remaining interval is closed.